

Sample Problem 5/16

At the instant represented, the disk with the radial slot is rotating about O with a counterclockwise angular velocity of 4 rad/sec which is decreasing at the rate of 10 rad/sec². The motion of slider A is separately controlled, and at this instant, $r = 6$ in., $\dot{r} = 5$ in./sec, and $\ddot{r} = 81$ in./sec². Determine the absolute velocity and acceleration of A for this position.

Solution. We have motion relative to a rotating path, so that a rotating coordinate system with origin at O is indicated. We attach x - y axes to the disk and use the unit vectors $\mathbf{i}$ and $\mathbf{j}$.

Velocity. With the origin at O , the term $\mathbf{v}_B$ of Eq. 5/12 disappears and we have

$$\textcircled{1} \quad \mathbf{v}_A = \boldsymbol{\omega} \times \mathbf{r} + \mathbf{v}_{\text{rel}}$$

- $\textcircled{2}$ The angular velocity as a vector is $\boldsymbol{\omega} = 4\mathbf{k}$ rad/sec, where $\mathbf{k}$ is the unit vector normal to the x - y plane in the $+z$ -direction. Our relative-velocity equation becomes

$$\mathbf{v}_A = 4\mathbf{k} \times 6\mathbf{i} + 5\mathbf{i} = 24\mathbf{j} + 5\mathbf{i} \text{ in./sec} \quad \text{Ans.}$$

in the direction indicated and has the magnitude

$$v_A = \sqrt{(24)^2 + (5)^2} = 24.5 \text{ in./sec} \quad \text{Ans.}$$

Acceleration. Equation 5/14 written for zero acceleration of the origin of the rotating coordinate system is

$$\mathbf{a}_A = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + \dot{\boldsymbol{\omega}} \times \mathbf{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}}$$

The terms become

$$\textcircled{3} \quad \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) = 4\mathbf{k} \times (4\mathbf{k} \times 6\mathbf{i}) = 4\mathbf{k} \times 24\mathbf{j} = -96\mathbf{i} \text{ in./sec}^2$$

$$\dot{\boldsymbol{\omega}} \times \mathbf{r} = -10\mathbf{k} \times 6\mathbf{i} = -60\mathbf{j} \text{ in./sec}^2$$

$$2\boldsymbol{\omega} \times \mathbf{v}_{\text{rel}} = 2(4\mathbf{k}) \times 5\mathbf{i} = 40\mathbf{j} \text{ in./sec}^2$$

$$\mathbf{a}_{\text{rel}} = 81\mathbf{i} \text{ in./sec}^2$$

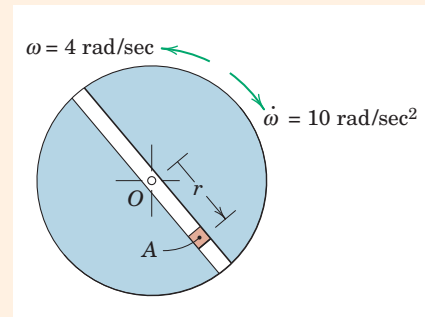
The total acceleration is, therefore,

$$\mathbf{a}_A = (81 - 96)\mathbf{i} + (40 - 60)\mathbf{j} = -15\mathbf{i} - 20\mathbf{j} \text{ in./sec}^2 \quad \text{Ans.}$$

in the direction indicated and has the magnitude

$$a_A = \sqrt{(15)^2 + (20)^2} = 25 \text{ in./sec}^2 \quad \text{Ans.}$$

Vector notation is certainly not essential to the solution of this problem. The student should be able to work out the steps with scalar notation just as easily. The correct direction of the Coriolis-acceleration term can always be found by the direction in which the head of the $\mathbf{v}_{\text{rel}}$ vector would move if rotated about its tail in the sense of $\boldsymbol{\omega}$ as shown.



Helpful Hints

- $\textcircled{1}$ This equation is the same as $\mathbf{v}_A = \mathbf{v}_P + \mathbf{v}_{A/P}$, where P is a point attached to the disk coincident with A at this instant.
- $\textcircled{2}$ Note that the x - y - z axes chosen constitute a right-handed system.
- $\textcircled{3}$ Be sure to recognize that $\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$ and $\dot{\boldsymbol{\omega}} \times \mathbf{r}$ represent the normal and tangential components of acceleration of a point P on the disk coincident with A . This description becomes that of Eq. 5/14b.

